Ex:-7:- 72, 97, 133, 182, 246, 327, ____?

72, 97, 133, 182, 246, ~~280~~, 327, [427]

$+25=5^2$ $+36=6^2$ $+49=7^2$ $+64=8^2$ $+81=9^2$ $+10^2=100$

logic:- Square difference series.

↳ In the same way we can go through 'n' different difference series by practicing.

(iii) Product series:- [× [Multiplication], ÷ [Division]]:-

Ex:-8:- 2, 4, 12, 48, 240, 1440, 10080, ____?

Ans:- 2, 4, 12, 48, 240, 1440, 10080, [80640]

×2 ×3 ×4 ×5 ×6 ×7 ×8

logic:- General product series.

Ex:-9:- 14, 42, 210, 1470, 16170, ____?

14, 42, 210, 1470, 16170, [210210]

×3 ×5 ×7 Not ×9 ×11 ×13 [prime No.s]

logic:- Prime product series:-

Ex:-10:- 12, 6, 6, 9, 18, 45, 135, ____?

12, 6, 6, 9, 18, 45, 135, [472.5]

$\times\frac{1}{2}$ $\times1=2\times\frac{1}{2}$ $\times1\frac{1}{2}=3\times\frac{1}{2}$ $\times2=4\times\frac{1}{2}$ $\times2\frac{1}{2}=5\times\frac{1}{2}$ $\times3=6\times\frac{1}{2}$ $7\times\frac{1}{2}=\times3\frac{1}{2}$

logic:- table product series [$\frac{1}{2}$ table here].

✱ Ex:-11:- 678, 336, 54, 20, ____?

Ans:- This logic is not looks any type of product but it's product series only.

If we observe clearly we can understand that

678 => 6X7X8 = 336

336 => 3X3X6 = 54

54 => 5X4 = 20

20 => 2X0 = [0] Ps our answer.

Logic:- Product of digits / digits Product series.

★ (iv) Combination Series:-

Till now we have seen +, −, X & ÷. But this Ps Combination of any 2 arithmetic operations of 4.

Ex:-12:- Jam taking an example combination as (+, X) with starting digit as 7. We can form 'n' number of seqieses by taking starting digit as same for all.

Ex:-12:- 7, 8, 20, 69, 292, ? = 1485

+1X1 +2X2 +3X3 +4X4 +5X5

Ex:-13:- 7, 8, 9, 10, 11, 12

+1X1 +1X1 +1X1 +1X1 +1X1

Ex:-14:- 7, 8, 10, 13, 17, 22

+1X1 +2X1 +3X1 +4X1 +5X1

Ex:-15:- 7, 8, 18, 57, 232, 1165

+1X1 +1X2 +1X3 +1X4 +1X5

Ex:-16:- 7, 8, 11, 16, 23, 31

+1X1 +1X3 +1X5 +1X7 +1X8

Ex:-17:- 7, 8, 33, 190, 1379, 12492

+1X1 +3X3 +5X5 +7X7 +9X9

(V) Square series:- When comes to square series we must know squares from 1 to 30. Then only we can able to do all the problems under square series.

Ex:18:- 125, 148, 173, 200, 229, ?.

Ans:- here in the question all the numbers are looks like square+4,
$121+4 = 11^2+4$, $144+4 = 12^2+4$, $169+4 = 13^2+4$, $196+4 = 14^2+4$, $225+4 = 15^2+4$
$\Rightarrow \therefore$ next one $16^2+4 = 256+4 = \boxed{260}$.

Logic:- n^2 + constant number [n^2 = square of a number]

Ex:-19:- 145, 171, 199, 229, 261, ?.

Ans:- here numbers are looks $145[12^2+1]$, $171[13^2+2]$, $199[14^2+3]$, $229[15^2+4]$, $261[16^2+5]$ => Squares are added by natural numbers. $\therefore$ next number = $17^2+6 = \boxed{295}[289+6]$

Logic:- n^2 + natural number.

* Ex:-20:- 22, 49, 156, 769, 5104, ?

Ans:- here this question looks little tought because the gap b/w every consecutive observation is increasing with larger gap, when this kind of scenarios comes in to picture mostly the logic is [squares + squares/cubes/ !] might be the logic.

So the above observations like.

$22 = 16+6 = 4^2+3!$, $49 = 25+24 = 5^2+4!$, $156 = 36+120 = 6^2+5!$

$769 = 49+720 = 7^2+6!$, $5104 = 64+5040 = 8^2+7!$

$\therefore$ the next number is $9^2+8! = 81+40,320 = \boxed{40,401}$